

Kaidi Zha

UNDERGRADUATE RESEARCHER · AI & MACHINE LEARNING

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“Building intelligent systems that see, learn, and act.”

Research Interests

Generative Models, Inference Acceleration

Dedicated to making large generative models faster and cheaper to deploy through **quantization and mixed-precision inference**, **efficient architecture design**, and **data-centric strategies**, with a focus on **inference throughput**, **memory efficiency**, and **training efficiency**.

Education

Tsinghua University

Beijing, China

B.S. IN MATHEMATICS AND PHYSICS

Sept. 2023 - Present (Expected 2027)

- Overall GPA: 3.6/4.0, with a steady upward trend — 3.9/4.0 in the most recent semester (Spring 2026).
- Earned A (4.0) in Pattern Recognition and Machine Learning, Reinforcement Learning and Control, and Fundamentals of Computer Graphics.

RELEVANT COURSEWORK

AI & ML

Pattern Recognition and Machine Learning (A), Reinforcement Learning and Control (A), Fundamentals of Computer Graphics (A), Statistical Machine Learning (B+)

Mathematics

Probability and Statistics (A), Experiments in Mathematics (A), Applied Stochastic Processes (A-), Complex Analysis (B+)

Computer Science

Data Structure and Algorithm (A-), Programming Fundamentals (A-), Modern Operating Systems (P), C++ Programming for Linux (P)

Research Experience

Q-DiT: Accurate Post-Training Quantization for Diffusion Transformers

Supervised by Prof. Yuan Meng,
Dept. of CS & Tech, Tsinghua
University

RESEARCH ASSISTANT

Jun. 2025 - Present

- Motivation:** Uniform-precision PTQ wastes the bit budget on diffusion Transformers, whose quantization sensitivity varies sharply across denoising timesteps and network layers — bits should be spent where the loss surface is steepest.
- Method:** Co-designed a timestep-aware, layer-wise mixed-precision PTQ framework grounded in second-order analysis: the Hessian trace of the denoising loss scores each timestep’s sensitivity, GPTQ (Optimal Brain Surgeon) error bounds per-layer weight perturbations, and Fisher information ranks each layer’s contribution to the training objective. Their combination turns bit allocation into a constrained optimization problem, solved exactly by dynamic programming under a global W4A4 budget, with a cross-adjustment mechanism that rebalances weight and activation bit-widths.
- System & Results:** Independently refactored the codebase into *qdiffusers*, a modular YAML-driven quantization–evaluation pipeline, and adapted it to **7 DiT-family models** (DiT-XL/2, FLUX.2, PixArt-Σ, OpenSora, Qwen-Image, HunyuanVideo, Wan2.2) with multi-GPU parallel GPTQ. Led all ImageNet-1K experiments — W4A4 achieves near-lossless FID 7.76 on DiT-XL/2 256×256; on FLUX.2-klein-4B, W4A4 FID 25.22 even **surpasses the FP16 baseline** (26.08).

ArchEGraph: A Large-Scale Graph Dataset for Geometry–Topology–Physics Aligned Building Energy Modeling

Supervised by Prof. Borong Lin,
School of Architecture, Tsinghua
Univ.; with UC Berkeley & MIT

RESEARCH ASSISTANT

Apr. 2026 - Present

- Role:** Second author of the manuscript; independently designed and executed the benchmark experiments for a dataset of **5,481 buildings and 49,326 validated simulation cases** ($>1.33 \times 10^5$ space nodes, 1.44×10^6 face nodes).
- Contributions:** Built the evaluation protocols for two benchmark tasks — reconstructing zone topology from polygonal meshes (M2G) and topology-informed zone-level load prediction — and benchmarked 7 model families (XGBoost, DeepSets, Set Transformer, Perceiver, GATv2, TransformerConv, GraphGPS) under cross-building and cross-climate splits.
- Findings:** Showed that graph-based surrogates learn the geometry–topology–physics coupling well in-distribution, while cross-climate and cross-layout generalization remains the key bottleneck — empirically validating the dataset’s design goal of exposing geometric and climatic shift.

Selected Projects

TrainYourOwnGPT: Building & Analyzing GPT-Style Language Models from Scratch

Pattern Recognition & Machine Learning, Course Project

TEAM OF 2; LED SCALING-LAW AND FINE-TUNING STUDIES

Spring 2026

- Implemented a GPT-style decoder (RoPE, SwiGLU, pre-norm blocks) from scratch in PyTorch, reaching **105.15 validation PPL** on Penn Treebank; ran a 5-configuration hyperparameter sweep over 3 seeds.
- Conducted a scaling-law study across 0.77M–5.46M parameters and controlled ablations of QK normalization, attention gating, and value embeddings.
- Fine-tuned Qwen2.5-0.5B with LoRA (0.89% trainable parameters) on arXiv abstracts with a pre-registered quantitative evaluation of format shift; compared autoregressive vs. masked-diffusion LMs (MDLM, LLaDA), including A100 throughput/latency benchmarks.

fair-SPLIT: Near-Optimal Decision Trees for Accuracy, Sparsity, and Fairness

Statistical Machine Learning, Course Project

TEAM OF 2; DEVELOPED THE SPLIT-FAMILY METHODS AND EXPERIMENTAL DESIGN

Spring 2026

- Motivation:** Greedy tree induction (CART) is myopic and offers no fairness guarantee, while simply dropping sensitive attributes does not remove indirect discrimination — we studied how far near-optimal trees plus lightweight post-processing can close this gap.
- Method & Experiments:** Implemented the SPLIT family, which replaces greedy induction with dynamic-programming lookahead that provably optimizes the most discriminative top layers before greedy completion; benchmarked against CART on **12 public datasets** (6 with sensitive attributes) × 5 seeds, measuring accuracy, training time, and SPD/DI/EOD fairness metrics — SPLIT variants win on COMPAS, HELOC, Heart, and Law School.
- Fairness:** Proposed Leaf-Pareto Fair Recalibration (LPFR), a deployable leaf-level post-processing method; label-aware calibration cuts the demographic-parity gap from 0.17 to **0.003** on Adult. Open-sourced at github.com/illusionsin30/fair-SPLIT.

AIDER: Taming Aleatoric Impulse in Off-Policy Reinforcement Learning

Reinforcement Learning & Control, Course Project

TEAM OF 8; EXPLORED DIFFUSION/FLOW-BASED MULTIMODAL POLICY EXTENSION

Spring 2026

- Team built DEAR, an EMA-anchored evidential regression operator for end-to-end aleatoric uncertainty estimation, and AIDER, which damps value-overestimation impulses without heuristic annealing — best average performance on the DeepMind Control Suite among 7 model-free baselines.
- Deployed on a 4WD vehicle longitudinal speed-tracking task with **RMSE 0.349 m/s** (~1.5% tracking error); personally explored integrating the framework with diffusion / normalizing-flow policies for multimodal action distributions.

CPU Path Tracer: A Physically-Based Renderer in C++

Fundamentals of Computer Graphics, Course Project

INDIVIDUAL PROJECT

Spring 2026

- Built a renderer from scratch: Whitted ray tracing and Monte Carlo path tracing with next-event estimation, Cook–Torrance glossy BRDF, Schlick Fresnel, depth of field, and BVH acceleration.
- Engineered deterministic lock-free multithreading (**std:::thread**) with a **49× speedup on 64 threads** and byte-identical output to the serial renderer.

Skills

Programming	Python, C/C++, CUDA, 汉语
Deep Learning	PyTorch, JAX, HuggingFace Transformers & Diffusers, PEFT/LoRA, quantization toolchains
Tools	Git, Linux, Docker, multi-GPU training, OpenMP / std:::thread
Languages	Chinese (Native), English (CET-4: 596, CET-6: 534)